

8th-Grade

TOPICS

1. Real Numbers
    - Reviews
    - Powers and Radicals
2. Algebraic Expressions
    - Definition of Algebraic Expressions
    - Identities and Factorization
    - Algebraic Ratios
3. Sequences and Functions
    - Sequences
    - Functions
    - Special Functions
4. Equations and Inequalities
    - Equations
    - System of Equations
    - Inequalities
5. Quadratic Equations
6. Probability and Statistics
    - Probability
    - Statistics
7. Mathematical Logic
8. Equations and Inequalities
    - Basic Geometric Concepts
    - Angle and Similarity in Triangles
    - Right Triangles and Trigonometry
    - Quadrilaterals

8th-Grade



1. Real Numbers
    - Powers and Radicals

1.2 - POWERS AND RADICAL NUMBERS

1. Understanding Square Roots(POWERS)

Definition

If $a^2 = b$ then a is the **square root** of b $(a \geq 0, b \geq 0)$.

We use the symbol $\sqrt{\ }$ to denote the square root of a number. $\sqrt{b}$ is read as 'the square root of b'. So if $a^2 = b$ then $a = \sqrt{b}$ $(a \geq b, b \geq 0)$.

1.2 - POWERS AND RADICAL NUMBERS

1. Understanding Square Roots(POWERS)

Here are the square roots of all the perfect squares from 1 to 100.

$1^2 = 1 \Rightarrow \sqrt{1} = 1$

$2^2 = 4 \Rightarrow \sqrt{4} = 2$

$3^2 = 9 \Rightarrow \sqrt{9} = 3$

$4^2 = 16 \Rightarrow \sqrt{16} = 4$

$5^2 = 25 \Rightarrow \sqrt{25} = 5$

$6^2 = 36 \Rightarrow \sqrt{36} = 6$

$7^2 = 49 \Rightarrow \sqrt{49} = 7$

$8^2 = 64 \Rightarrow \sqrt{64} = 8$

$9^2 = 81 \Rightarrow \sqrt{81} = 9$

$10^2 = 100 \Rightarrow \sqrt{100} = 10$

1.2 - POWERS AND RADICAL NUMBERS

1. Understanding Square Roots(POWERS)

EXAMPLE 1 Evaluate each square root.

a. $\sqrt{81}$ b. $\sqrt{1}$ c. $\sqrt{0}$ d. $\sqrt{64}$ e. $\sqrt{9}$ f. $\sqrt{0.64}$

g. $-\sqrt{100}$ h. $-\sqrt{0.09}$ i. $\sqrt{-4}$ j. $\sqrt{(-4)^2}$ k. $\sqrt{-4^2}$

8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

1. Understanding Square Roots(POWERS)

Rule

If $x \in R$ then

$$\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x \geq 0. \\ -x & \text{if } x < 0. \end{cases}$$

In other words, if x is a non-negative real number, then $\sqrt{x^2} = x$, and

if x is a negative real number, then $\sqrt{x^2} = -x$.

1.2 - POWERS AND RADICAL NUMBERS

1. Understanding Square Roots(POWERS)

EXAMPLE 2 Evaluate each square root.

a. $\sqrt{100}$ b. $\sqrt{121}$ c. $\sqrt{144}$ d. $\sqrt{169}$ e. $\sqrt{225}$ f. $\sqrt{361}$

g. $\sqrt{400}$ h. $\sqrt{625}$ i. $\sqrt{1225}$ j. $\sqrt{10000}$

8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

2. Properties of Square Roots

Property

For any real number a and b, where $a \geq 0$, and $b \geq 0$,

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{a \cdot b}.$$

For example,

$$\sqrt{25 \cdot 16} = \sqrt{25} \cdot \sqrt{16} = 5 \cdot 4 = 20,$$

$$\sqrt{3} \cdot \sqrt{27} = \sqrt{3 \cdot 27} = \sqrt{81} = 9,$$

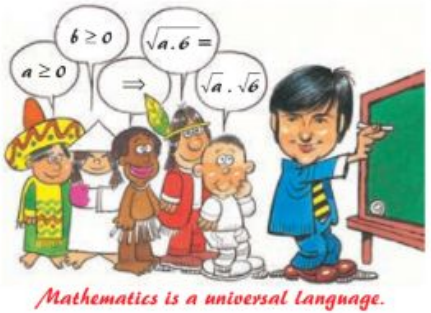
$$\sqrt{36a^2} = \sqrt{36}\sqrt{a^2} = 6a \ (a \geq 0), \text{ and}$$

$$\sqrt{5} \cdot \sqrt{5} = \sqrt{5 \cdot 5} = \sqrt{25} = 5.$$

Note

If $a \geq 0$ then

$$\sqrt{a} \cdot \sqrt{a} = \sqrt{a \cdot a} = \sqrt{a^2} = a.$$



8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

2. Properties of Square Roots

EXAMPLE 3 Simplify each of the following.

a. $\sqrt{2} \cdot \sqrt{8}$ b. $\sqrt{7} \cdot \sqrt{7}$ c. $\sqrt{50} \cdot \sqrt{2}$ d. $\sqrt{25} \cdot \sqrt{1}$ e. $\sqrt{576}$ f. $\sqrt{10} \cdot \sqrt{90}$

1.2 - POWERS AND RADICAL NUMBERS

2. Properties of Square Roots

Property

For any real numbers a and b, where $a \geq 0$, and $b > 0$,

$$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}.$$

If $a > 0$ then

$$\frac{\sqrt{a}}{\sqrt{a}} = \sqrt{\frac{a}{a}} = \sqrt{1} = 1.$$

For example,

$$\frac{\sqrt{24}}{\sqrt{6}} = \sqrt{\frac{24}{6}} = \sqrt{4} = 2, \text{ and}$$

$$\sqrt{\frac{1}{49}} = \frac{\sqrt{1}}{\sqrt{49}} = \frac{1}{7}.$$

1.2 - POWERS AND RADICAL NUMBERS

2. Properties of Square Roots

EXAMPLE 4 Simplify the expressions.

a. $\sqrt{\frac{25}{9}}$ b. $\frac{\sqrt{50}}{\sqrt{2}}$ c. $\sqrt{\frac{16}{49}}$ d. $\sqrt{\frac{1}{64}}$ e. $-\sqrt{\frac{1}{100}}$ f. $\sqrt{\frac{625}{144}}$

g. $\frac{\sqrt{24a^3}}{\sqrt{6a}}$ h. $\frac{\sqrt{a^5 \cdot b^6}}{\sqrt{ab^2}}$ i. $\frac{\sqrt{x \cdot y}}{\sqrt{x^3 \cdot y^3}}$

1.2 - POWERS AND RADICAL NUMBERS

2. Properties of Square Roots

Property

For any real number a and $n \in Z$, $(\sqrt{a})^n = \sqrt{a^n} \quad (a \geq 0)$.

Proof

$$(\sqrt{a})^n = \underbrace{\sqrt{a} \cdot \sqrt{a} \cdot \sqrt{a} \cdot \ldots \cdot \sqrt{a}}_{n \text{ factors of } \sqrt{a}} = \underbrace{\sqrt{a \cdot a \cdot a \cdot \ldots \cdot a}}_{n \text{ factors of } a} = \sqrt{a^n}$$

For example, $(\sqrt{a})^2 = \sqrt{a^2} = a$,

$(\sqrt{5})^3 = \sqrt{5^3} = \sqrt{125}$, *and*

$(\sqrt{2})^8 = \sqrt{2^8} = \sqrt{256} = 16$.

1.2 - POWERS AND RADICAL NUMBERS

2. Properties of Square Roots

EXAMPLE 5 Evaluate

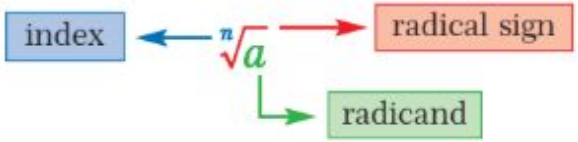
$$(\sqrt{2})^4 + (\sqrt{5})^4 - (\sqrt{5})^2 - (\sqrt{2})^6.$$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Definition

A **radical expression** is an expression of the form $\sqrt[n]{a}$.



Square roots have index 2. However, we usually write square roots in their shorter form, $\sqrt{a}$:

$$\sqrt[2]{a} = \sqrt{a}$$

8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Definition

A **mixed radical** is a radical of the form

$$x \cdot \sqrt[n]{a} \quad (x \in Q, \quad x \notin \{-1, 0, 1\})$$

For example, $3\sqrt{2}$, $6\sqrt{7}$, and $9\sqrt{115}$ are mixed radicals.

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Property

For any real numbers a and b, where $a \geq 0$ and $b \geq 0$,

$$\sqrt{a^2 \cdot b} = a\sqrt{b} \quad \text{and} \quad a\sqrt{b} = \sqrt{a^2 \cdot b}.$$

For example,

$$\sqrt{8} = \sqrt{4 \cdot 2} = \sqrt{2^2 \cdot 2} = \sqrt{2^2} \cdot \sqrt{2} = 2\sqrt{2},$$

$$\sqrt{27} = \sqrt{9 \cdot 3} = \sqrt{3^2 \cdot 3} = \sqrt{3^2} \cdot \sqrt{3} = 3\sqrt{3},$$

$$\sqrt{32} = \sqrt{16 \cdot 2} = \sqrt{4^2 \cdot 2} = \sqrt{4^2} \cdot \sqrt{2} = 4\sqrt{2}, \text{ and}$$

$$\sqrt{50} = \sqrt{25 \cdot 2} = \sqrt{5^2 \cdot 2} = \sqrt{5^2} \cdot \sqrt{2} = 5\sqrt{2}.$$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

EXAMPLE 6 Simplify the expressions.

a. $\sqrt{8} + 2\sqrt{32} - \sqrt{18} + \sqrt{72} - \sqrt{98}$

b. $2\sqrt{48} + 3\sqrt{27} - \sqrt{108} + \sqrt{243}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

EXAMPLE 7 Write the numbers as pure radicals.

a. $2\sqrt{2}$ b. $3\sqrt{5}$ c. $5\sqrt{3}$ d. $10\sqrt{10}$ e. $x\sqrt{y}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Property

For any non-zero real numbers a, b, c, and x,

$$a\sqrt{x} + b\sqrt{x} - c\sqrt{x} = (a + b - c)\sqrt{x}.$$

Note

$\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$

For example,

$\sqrt{9} + \sqrt{16} = 3 + 4 = 7$, but $\sqrt{9 + 16} = \sqrt{25} = 5$.

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

EXAMPLE 8 Perform the operations.

a. $\sqrt{3} + \sqrt{3}$ b. $2\sqrt{5} + \sqrt{5}$ c. $3\sqrt{6} + 4\sqrt{6}$ d. $10\sqrt{5} - 3\sqrt{5}$

e. $\sqrt{50} + \sqrt{98} + \sqrt{162}$ f. $5\sqrt{x} - \sqrt{9x} + \sqrt{64x}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

EXAMPLE 9 Compare the following numbers.

a. $\sqrt{7} \ldots 3$ b. $3\sqrt{5} \ldots 2\sqrt{10}$ c. $2\sqrt{7} \ldots 3\sqrt{3}$ d. $-2\sqrt{3} \ldots -3\sqrt{2}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Property

Let a, b, m, and n be four real numbers, satisfying $a = m + n$ and $b = m \cdot n$. Then,

1. $\sqrt{m} + \sqrt{n} = \sqrt{a + 2\sqrt{b}}$
2. $\sqrt{m} - \sqrt{n} = \sqrt{a - 2\sqrt{b}}$ $(m > n)$

EXAMPLE 10 Simplify the expressions. Use the property to help you.

a. $\sqrt{3 + 2\sqrt{2}}$ b. $\sqrt{5 + 2\sqrt{6}}$ c. $\sqrt{6 + \sqrt{32}}$ d. $\sqrt{6 - 4\sqrt{2}}$

e. $\sqrt{5 + \sqrt{21}}$ f. $\sqrt{2 + \sqrt{3}}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Check Yourself 1

1. Simplify the expressions.

   a. $\sqrt{2} \cdot \sqrt{2}$ b. $\sqrt{8} \cdot \sqrt{32}$ c. $\sqrt{3x} \cdot \sqrt{12x}$ d. $\sqrt{2} \cdot \sqrt{18}$

   e. $\dfrac{\sqrt{32}}{\sqrt{2}}$ f. $\dfrac{\sqrt{12}}{\sqrt{3}}$ g. $\dfrac{\sqrt{a} \cdot \sqrt{b}}{\sqrt{a^3 \cdot b^3}}$ h. $-\sqrt{\dfrac{1}{49}}$

2. Evaluate the following.

   a. $(\sqrt{3})^2 + (\sqrt{4})^4 - (\sqrt{5})^2 - (\sqrt{2})^4$ b. $(\sqrt{a})^4 + (\sqrt{b})^2 - (\sqrt{c})^6$

3. Simplify the expressions.

   a. $\sqrt{18}$ b. $\sqrt{50}$ c. $\sqrt{48}$ d. $\sqrt{20}$ e. $5\sqrt{3} - 2\sqrt{3} + \sqrt{3}$

   f. $2\sqrt{2} + 3\sqrt{2} - 4\sqrt{2}$ g. $\sqrt{50} - \sqrt{18} - \sqrt{32}$ h. $\sqrt{12x} + \sqrt{27x} - \sqrt{48x}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

Check Yourself 1

4. Write each number as a pure radical.

   a. $5\sqrt{3}$ b. $3\sqrt{5}$ c. $4\sqrt{2}$ d. $2\sqrt{5}$ e. $a\sqrt{b}$

5. Perform the operations.

   a. $6\sqrt{\frac{10}{3}}+2\sqrt{\frac{15}{2}}$ b. $5\sqrt{2}-\sqrt{8}$ c. $\sqrt{27}-\sqrt{48}$ d. $\sqrt{\frac{27}{4}}+\sqrt{\frac{75}{4}}-\sqrt{\frac{12}{4}}$

6. Compare the numbers.

   a. $3\sqrt{5}$ and $2\sqrt{10}$ b. $\sqrt{\frac{1}{2}}$ and $\sqrt{\frac{1}{3}}$ c. $-2\sqrt{5}$ and $-3\sqrt{3}$

7. Write each expression in its simplest form.

   a. $\sqrt{3-2\sqrt{2}}$ b. $\sqrt{6+2\sqrt{8}}$ c. $\sqrt{7-2\sqrt{10}}$ d. $\sqrt{3+\sqrt{8}}$ e. $\sqrt{9-4\sqrt{5}}$ f. $\sqrt{7-\sqrt{48}}$

   g. $(\sqrt{6}-\sqrt{2})\cdot(\sqrt{8+2\sqrt{12}})$ h. $(\sqrt{7}+1)\cdot(\sqrt{8-\sqrt{28}})$ i. $\sqrt{3-\sqrt{5}}\cdot\sqrt{3+\sqrt{5}}$

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

EXAMPLE 11 Simplify the following.

a. $\sqrt{4+\sqrt{21+\sqrt{13+\sqrt{9}}}}$

b. $\sqrt{6\sqrt{6\sqrt{72\sqrt{\frac{1}{16}\sqrt{16}}}}}$

c. $\sqrt{1+\sqrt{1+\frac{9}{16}}}$

8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

3. Working with Pure and Mixed Radicals

EXAMPLE 12

a. Evaluate $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\ \ldots}}}}$

b. $\sqrt{a\sqrt{a\sqrt{a\sqrt{a\ \ldots}}}} = 7$. Find a.

c. $\sqrt{x+\sqrt{x+\sqrt{x+\ldots}}} = 5$. Find x.

8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

4. Multiplying Square Roots

EXAMPLE 13 Perform the operations.

a. $\sqrt{2}(\sqrt{5} + \sqrt{3})$ b. $\sqrt{3}(3\sqrt{3} + 2\sqrt{2})$ c. $2\sqrt{5}(\sqrt{3} + \sqrt{2} + 2\sqrt{5} - \sqrt{7})$

1.2 - POWERS AND RADICAL NUMBERS

4. Multiplying Square Roots

EXAMPLE 14 Multiply and simplify.

a. $(\sqrt{2} + \sqrt{3}) \cdot (\sqrt{2} + \sqrt{3})$

b. $(5 + \sqrt{5}) \cdot (5 + \sqrt{5})$

1.2 - POWERS AND RADICAL NUMBERS

4. Multiplying Square Roots

EXAMPLE 15 Multiply and simplify.

a. $(\sqrt{2} + 1) \cdot (\sqrt{2} - 1)$

b. $(\sqrt{5} + \sqrt{3}) \cdot (\sqrt{5} - \sqrt{3})$

c. $(1 - 2\sqrt{2}) \cdot (1 + 2\sqrt{2})$

d. $(\sqrt{a} + 1) \cdot (\sqrt{a} - 1)$

e. $(\sqrt{a} + \sqrt{b}) \cdot (\sqrt{a} - \sqrt{b})$

1.2 - POWERS AND RADICAL NUMBERS

4. Multiplying Square Roots

EXAMPLE 16 Multiply and simplify.

a. $\sqrt{3+\sqrt{5}} \cdot \sqrt{3-\sqrt{5}}$ b. $\sqrt{2+\sqrt{2}} \cdot \sqrt{2-\sqrt{2}}$ c. $\sqrt{a+\sqrt{b}} \cdot \sqrt{a-\sqrt{b}}$

1.2 - POWERS AND RADICAL NUMBERS

4. Multiplying Square Roots

EXAMPLE 17 Multiply and simplify.

a. $(\sqrt{3} + \sqrt{2}) \cdot (\sqrt{5} - 1)$

b. $(\sqrt{5} + \sqrt{3}) \cdot (\sqrt{7} + \sqrt{2})$

c. $(2\sqrt{3} + 1) \cdot (\sqrt{5} + 1)$

d. $(3\sqrt{2} - 2) \cdot (\sqrt{5} - \sqrt{3})$

1.2 - POWERS AND RADICAL NUMBERS

5. Rationalizing Denominators

Definition: $\frac{\sqrt{a}}{\sqrt{b}} = \frac{\sqrt{a}}{\sqrt{b}} \cdot \frac{\sqrt{b}}{\sqrt{b}} = \frac{\sqrt{a \cdot b}}{\sqrt{b \cdot b}} = \frac{\sqrt{ab}}{b}$ **, b is not equal to 0**

Example:

$$\frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{3} \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \frac{\sqrt{3 \cdot 2}}{\sqrt{2 \cdot 2}} = \frac{\sqrt{6}}{\sqrt{4}} = \frac{\sqrt{6}}{2}$$

1.2 - POWERS AND RADICAL NUMBERS

5. Rationalizing Denominators

Remark

$(a + b)(a - b) = a^2 - b^2$

$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$ where $a \geq 0$ and $b \geq 0$.

EXAMPLE 18 Rationalize the denominators.

a. $\dfrac{\sqrt{5}}{3-2\sqrt{2}}$ b. $\dfrac{\sqrt{3}-\sqrt{2}}{2\sqrt{2}-1}$ c. $\dfrac{\sqrt{6}+\sqrt{2}}{1-\sqrt{3}}$ d. $\dfrac{3\sqrt{2}-2}{5+2\sqrt{5}}$

1.2 - POWERS AND RADICAL NUMBERS

5. Rationalizing Denominators

Remark

$(a + b)(a - b) = a^2 - b^2$

$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$ where $a \geq 0$ and $b \geq 0$.

EXAMPLE 19 Rationalize the denominators to find the sum.

$$\frac{\sqrt{3}}{\sqrt{3}+2\sqrt{2}} + \frac{\sqrt{2}}{\sqrt{3}-2\sqrt{2}}$$

8th-Grade

1.2 - POWERS AND RADICAL NUMBERS

Check Yourself 2

1. Rationalize the denominators and simplify.

a. $\sqrt{\frac{3}{7}}$ b. $\frac{\sqrt{5}}{\sqrt{3}}$ c. $2\sqrt{\frac{1}{2}}$ d. $-5 \cdot \frac{\sqrt{3}}{\sqrt{5}}$ e. $\frac{3\sqrt{5}}{\sqrt{3}}$ f. $\frac{3\sqrt{5}}{2\sqrt{6}}$

g. $\frac{1}{2\sqrt{10}}$ h. $\frac{\sqrt{a \cdot b}}{\sqrt{a^3}}$ i. $\frac{2\sqrt{x \cdot y}}{\sqrt{x^3 \cdot y^3}}$ j. $\frac{\sqrt{a^3 \cdot b^4}}{\sqrt{a \cdot b^2}}$

2. Rationalize the denominators and simplify.

a. $\frac{1}{\sqrt{2}-1}$ b. $\frac{\sqrt{2}}{\sqrt{3}-\sqrt{2}}$ c. $\frac{2}{\sqrt{5}+1}$ d. $\frac{3}{\sqrt{6}-2}$ e. $\frac{\sqrt{5}+1}{\sqrt{5}-1}$ f. $\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}-\sqrt{2}}$

g. $\frac{2\sqrt{5}-2}{2\sqrt{10}-6}$ h. $\frac{2\sqrt{5}-\sqrt{7}}{\sqrt{7}-\sqrt{5}}$ i. $\frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}$ j. $\frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}$ k. $\frac{3}{3\sqrt{3}-2\sqrt{7}}$

3. Rationalize the denominators and simplify.

a. $\frac{1}{\sqrt{5}+\sqrt{2}} + \frac{1}{\sqrt{5}-\sqrt{2}}$ b. $\frac{\sqrt{2}}{2-\sqrt{2}} + \frac{\sqrt{2}}{2+\sqrt{2}}$ c. $\frac{1}{\sqrt{3}+\sqrt{3}} \cdot \frac{1}{\sqrt{3}-\sqrt{3}}$

d. $\frac{2}{\sqrt{2}+1} - \frac{2}{\sqrt{2}-1} + \frac{3}{2\sqrt{2}}$ e. $\frac{3}{\sqrt{3}+1} - \frac{2}{1-\sqrt{3}} + \frac{3}{3\sqrt{3}}$

8th-Grade